F FWallSim · Firewall Course

CLASS 13

Dynamic routing: OSPF over VPN and BGP

OSPF neighbors, BGP peering, administrative distance

Advanced | ~30 minutes | FWallSim v1.1.0 (FortiOS 7.4 style)

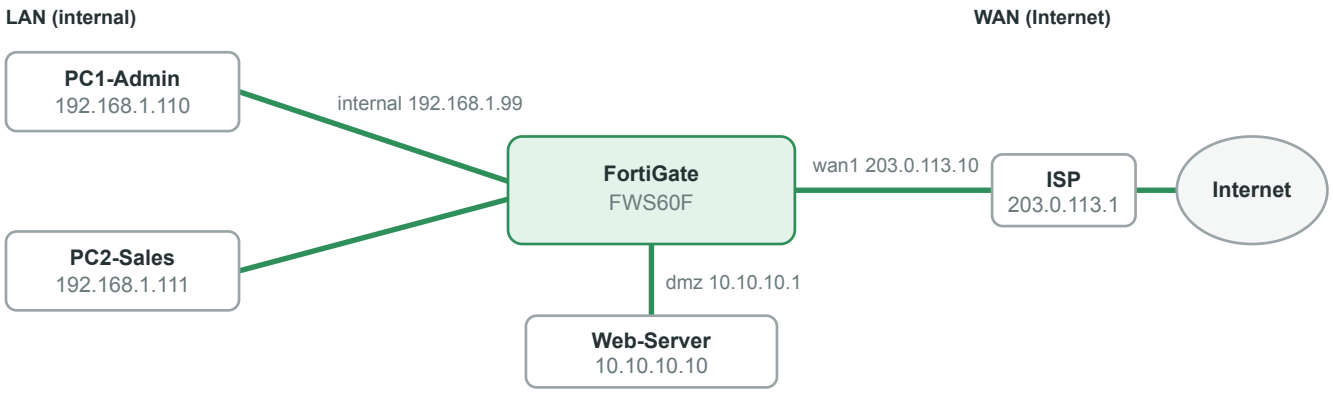
YOU WILL LEARN

- OSPF neighbors
- BGP peering
- administrative distance

BEFORE YOU START

Finish Class 12 first.

The lab network



The goal

THE IDEA

The VPN to the branch is already up with a static route. Replace the static route with OSPF over the tunnel, and peer with your ISP using BGP (details on Network > BGP and OSPF).

How to start

1. Open FWallSim and log in.
2. Click **Labs** in the header.
3. Find **Dynamic routing: OSPF over VPN and BGP** and click **Start**.
4. The Lab Guide panel opens. Each task below gets a green tick as soon as it works.

Tasks

1. **BGP session with the ISP is Established**

   Network > BGP: Local AS 65010, neighbor 203.0.113.1 remote AS 64500.

2. **OSPF neighbor 10.20.0.1 is Full**

   Give tunnel to-branch IP 169.254.1.1/32 with remote IP 169.254.1.2/30 (Network > Interfaces), then OSPF: area 0.0.0.0, network 169.254.1.0/30.

3. **10.20.0.0/24 is now learned by OSPF (static route removed)**

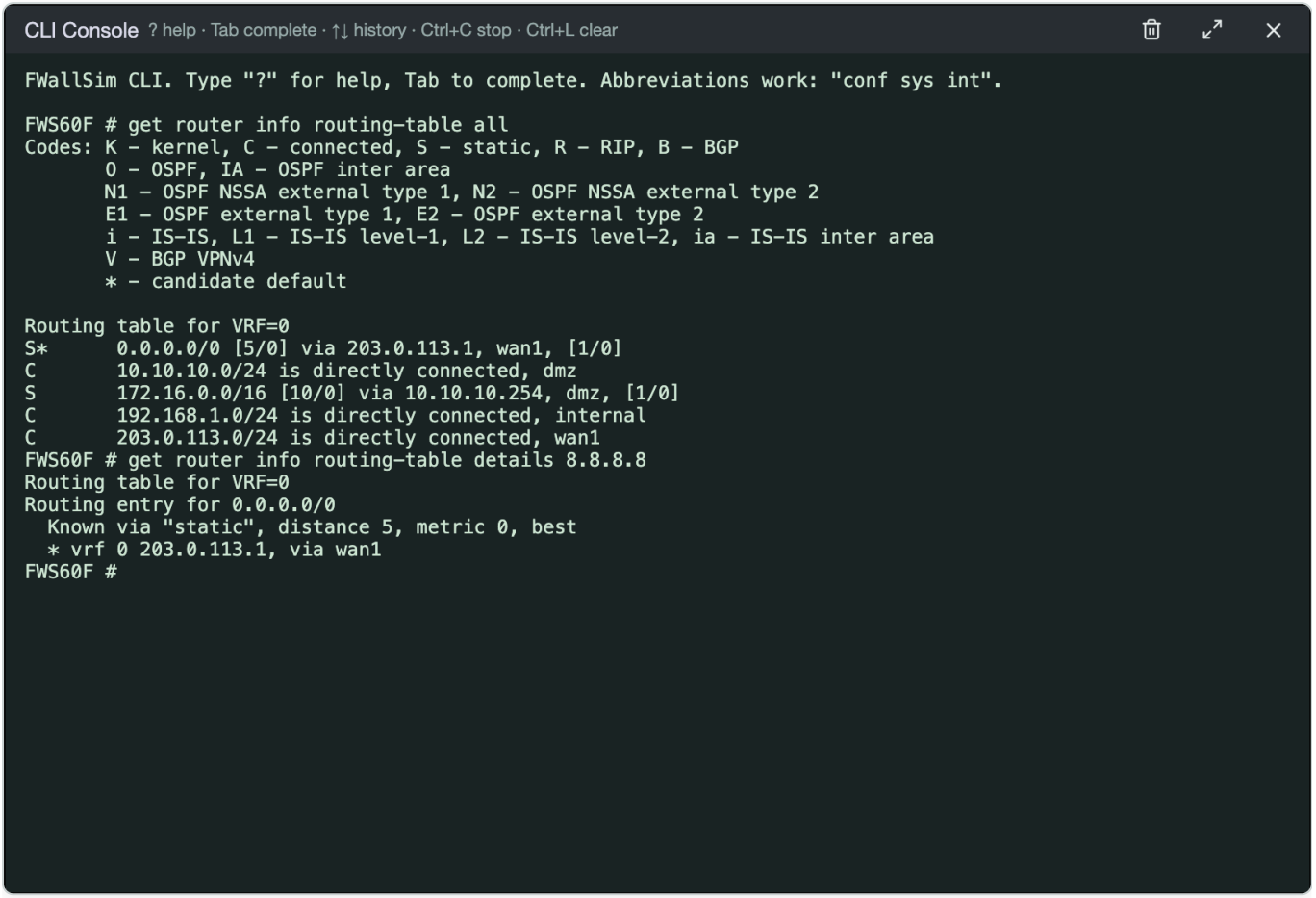
   Delete the static route to 10.20.0.0/24 via the tunnel. Static (distance 10) beats OSPF (110), so the OSPF route only becomes active without it. Keep the blackhole.

4. **PC1 still reaches Branch-PC**

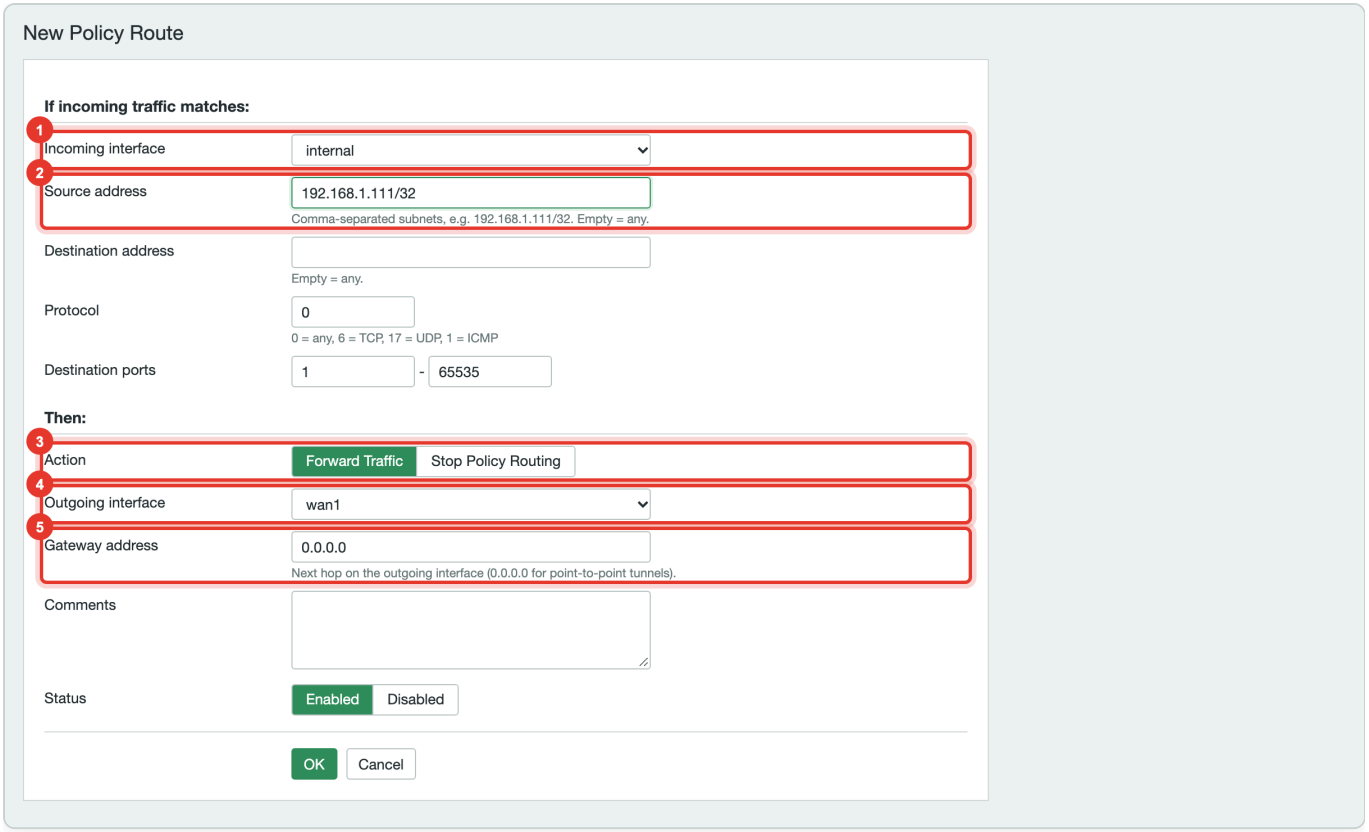
   After the static route is gone, send PC1-Admin → 10.20.0.10, Ping (Traffic Generator quick test "PC1 pings Branch-PC"). The Packet Walk shows the OSPF route.

Screenshots

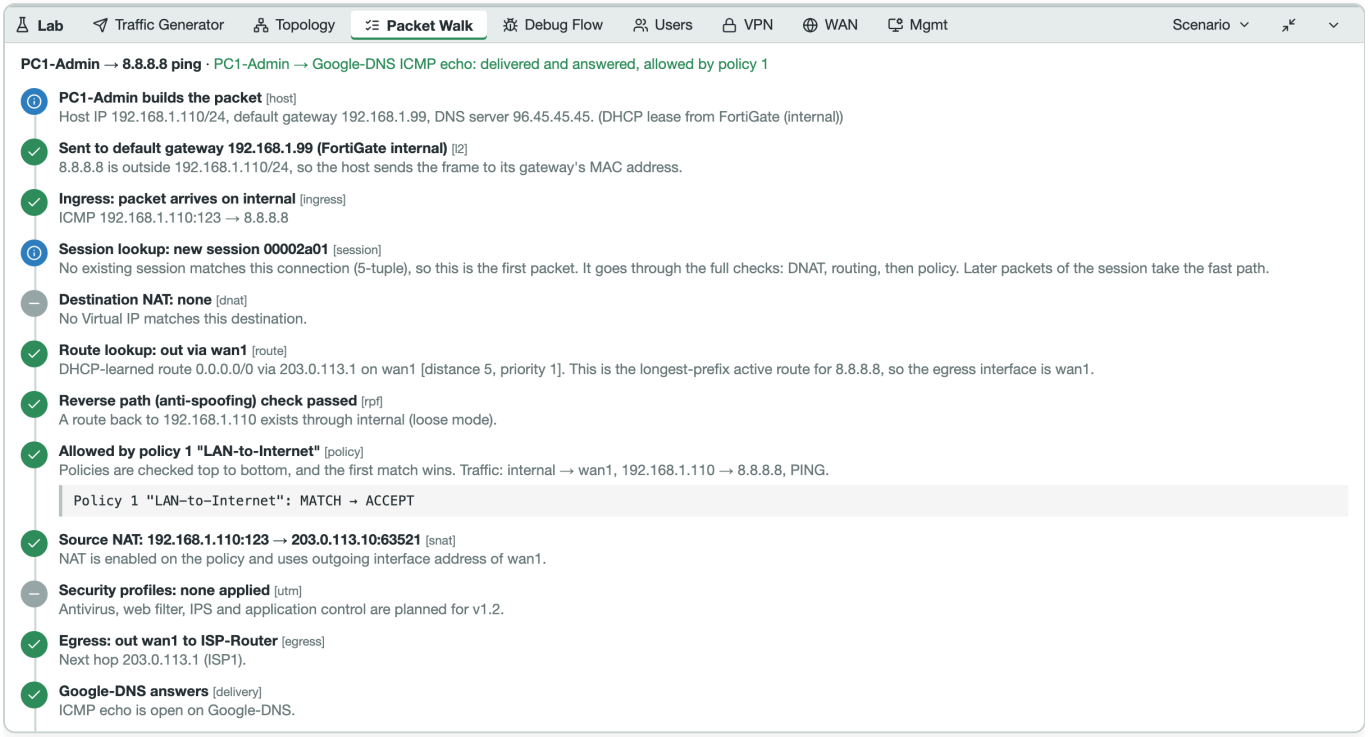
Numbered red markers show where to click or what to look at.



CLI Routing



Policy Route Form



Route Walk

Dashboard · Network

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Static & Dynamic Routing (equivalent of: get router info routing-table all / database)

Network	Gateway IP	Interfaces	Distance	Priority	Type	Status
10.10.10.0/24	0.0.0.0	dmz	0		C Connected	Active
192.168.1.0/24	0.0.0.0	internal	0		C Connected	Active
203.0.113.0/24	0.0.0.0	wan1	0		C Connected	Active
0.0.0.0/0	203.0.113.1	wan1	5	1	S* Static (DHCP)	Active

DHCP

Device	IP Address	Interface	Gateway / DNS handed out	Status
PC1-Admin	192.168.1.110	internal	192.168.1.99 / 96.45.45.45	Leased
PC2-Sales	192.168.1.111	internal	192.168.1.99 / 96.45.45.45	Leased

Interface traffic (since boot)

Interface	Link	IP	RX bytes	TX bytes	RX packets	TX packets
dmz	up	10.10.10.1/24	0 B	0 B	0	0
wan1	up	203.0.113.10/24	0 B	0 B	0	0
wan2	no link	–	0 B	0 B	0	0
internal	up	192.168.1.99/24	0 B	0 B	0	0

Lab | Traffic Generator | Topology | Packet Walk | Debug Flow | Users | VPN | WAN | Mgmt | Scenario

Routing Monitor

New Static Route

1 Destination: 172.16.0.0/16
Subnet: 0.0.0.0/0 is the default route

2 Interface: dmz

3 Gateway Address: 10.10.10.254
Must be inside the subnet of the chosen interface, or the route stays inactive.

4 Administrative Distance: 10

5 Priority: 1
Advanced: tie-breaker between routes with the same distance (lower wins).

Comments:

Status: Enabled | Disabled

OK | Cancel

Static Form

Static Routes

+ Create New | Edit | Clone | Delete | Search

Destination	Gateway IP	Interface	Distance	Priority	Status	Comments
172.16.0.0/16	10.10.10.254	dmz	10	1	Active	
172.16.0.0/16		Blackhole	254	1	Inactive: higher distance than another route to the same prefix	

The default route learned by DHCP on wan1 does not appear here. See it under Dashboard > Network > Routing. Lower distance wins for the same prefix; for equal distance, lower priority wins.

Static List

Next: Class 14: Nobody can browse the internet